PIC Application Note
O-band HPSOA Usage

12-14-2022

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1. Revision History

Rev	Date	Revised by	Comments
1.0	12/12/2022	H. Zhao	Initial draft
2.0	12/14/2022	H. Zhao	Update document ID

2. References

[1] L. Coldren, S. Q. Corzine, M. L. Mashanovitch, "Diode lasers and photonic integrated circuits," John Wiley & Sons, Hoboken, 2nd ed, 2011.

[2] R. Lennox et al., "Impact of bias current distribution on the noise figure and power saturation of a multicontact semiconductor optical amplifier," Optics Lett., vol.36, pp. 2521-2523, 2011.

3. Glossary

HPSOA High Power Semiconductor Optical Amplifier

SOI Silicon on Insulator substrates

WPE Wall-Plug Efficiency

NF Noise Figure

4. Document Scope

This Application Note provides operation conditions and performance for OpenLight O-band high power Semiconductor Optical Amplifier (HPSOA).

5. Introduction

Semiconductor optical amplifiers (SOAs) are designed to amplify optical signal through group III-V gain materials. High power semiconductor optical amplifiers (HPSOAs) on a hybrid platform enable >100mW optical power per waveguide with high wall-plug efficiency. As shown in Fig. 1, the III-V gain material is heterogeneously bonded onto Silicon on Insulator (SOI) substrates. Light is guided in and out of the SOA vias evanescent coupling by the Si waveguides underneath the gain material.

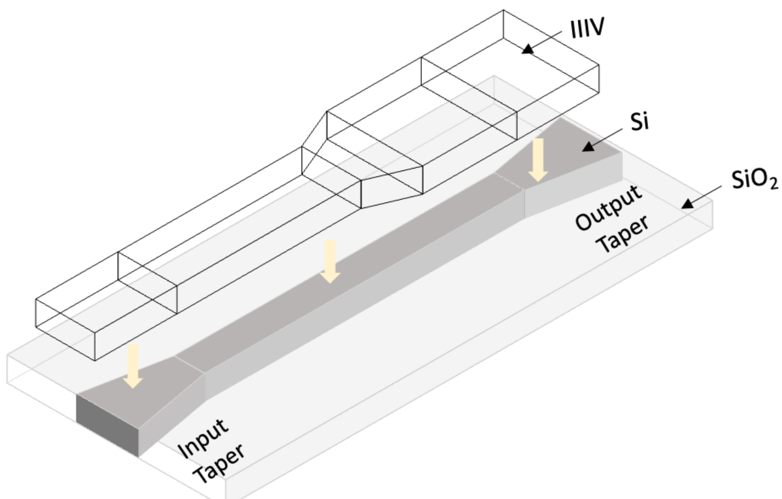

Figure 1. Schematic of HPSOA design.

The HPSOA described in this Application Note is designed for O-band operation (centered at 1311 nm). The III-V gain region is a multi-width structure, the detail of this design is discussed in Sec.6. For general use purpose, a lookup table for different operation conditions (temperature, current and wavelength) is covered in Sec. 7.

6. HPSOA Design

6.1. Geometry of the HPSOA Design

Output saturation power of an SOA (P_{os}) is given by:

$$P_{os} = \frac{g_o ln2}{g_o - 2} \frac{Wdh\nu}{a\Gamma\tau} \quad \text{(Eq.1)}$$

where W and d are the width and thickness of the active layer, a is the differential gain of the materials, Γ is the confinement factor, τ is the carrier lifetime, h is the Planck constant and ν is the optical frequency. To achieve a high P_{os}, the III-V epi layers are tailored for a low confinement factor and a multi-section SOA with flared width towards output is designed.

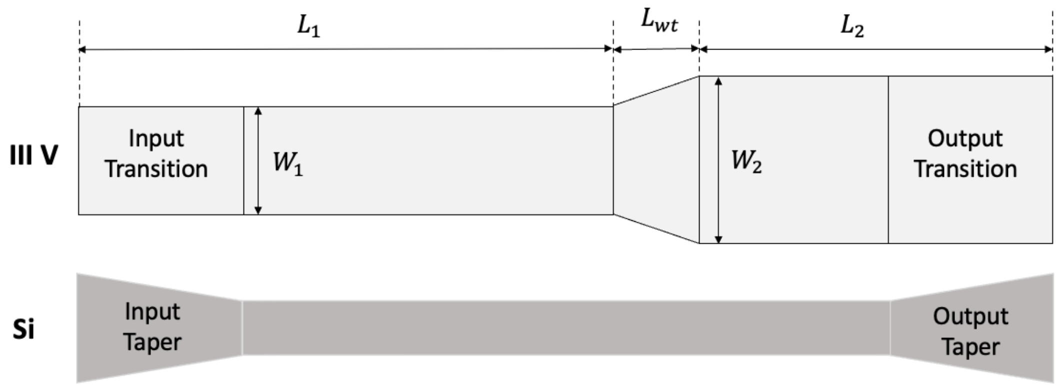

Figure 2. Top view of the HPSOA design.

Figure 2 shows the top view of both III-V gain and the underlying Si waveguides. The III-V gain contains three sections: narrow-ridge input (L_1), transition taper (L_{wt}) and wide-ridge output (L_2). The size of each section is demonstrated in Table 1.

Table 1. Dimension of the HPSOA.

W_1 [um]	L_1 [um]	L_{wt} [um]	W_2 [um]	L_2 [um]
2	1260	60	4	400

6.2. HPSOA Performance

The HPSOA design can be considered as a chain of two SOAs of different widths (W_1 and W_2), so the total NF is given by:

$$NF = NF_1 + \frac{NF_2 - 1}{G_1} \quad \text{(Eq. 2)}$$

Wall-plug efficiency (WPE) is defined as the ratio of output optical power (P_o) and electrical power consumed:

$$WPE = \frac{P_o}{V \times I} \quad \text{(Eq. 3)}$$

where V is the voltage, I is the current.

SOA is a P-I-N diode structure. For a single diode, the I-V characteristic is described by:

$$I = I_0 \exp\left[\frac{q}{nkT}(V - IR_s)\right] \quad \text{(Eq.4)}$$

where I_0 is the saturation current, n is the diode ideality, k is the Boltzmann constant, Rs is the series resistance, T is the temperature, and q is the unit charge. The equation can be further written in the log scale as:

$$V = V_{turn-on} + IR_s = \frac{nkT}{q}[(Ln(I) - Ln(I_0))] + IR_s \quad \text{(Eq.5)}$$

The turn on voltage is about 1.05 V. Series resistance is estimated by a response surface model:

$$R_s = \frac{4.34}{W\,[um]} + \frac{2151}{L\,[um]} - 0.992 \quad \text{(Eq.6)}$$

where W is the SOA ridge width and L is the length of the diode.

Based on the above diode IV model, a simplified circuit, as shown in Fig. 3., was used to solve the voltage and current that is needed to drive the HPSOA. R_{s1} and R_{s2} are the series resistance for the input and output section of the HPSOA.

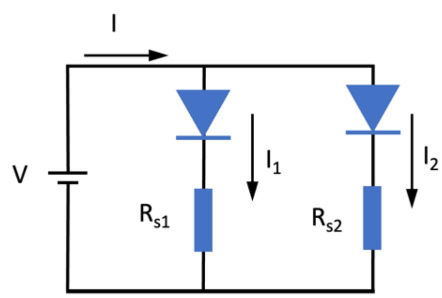

Figure 3. Equivalent circuit for HPSOA.

$$I = I_1 + I_2$$

$$I_1 = (V - V_{turn-on})/R_{s1}$$

$$I_2 = (V - V_{turn-on})/R_{s2}$$

$$V_{turn-on} = 1.05\,[V], R_{s1} = 2.89\,[ohm],\ R_{s2} = 5.47\,[ohm]$$

7. Performance Summary

Table 2 lists the performance of the HPSOA under the following operation conditions:

T = [35, 55, 70, 80] °C

I = [0.18, 0.23, 0.28] A

$\lambda = [1304, 1311, 1318]$ nm

Table 2. HPSOA Performance Summary.

Index	T [°C]	I [A]	λ [nm]	Pin [dBm]	Pout [dBm]	WPE [%]	NF [dB]
1	35	0.18	1304	10	19.4	35.1	9.72
2	35	0.18	1311	10	18.8	30.6	9.31
3	35	0.18	1318	10	19	32.1	8.11
4	35	0.23	1304	10	20.8	34.9	9.68
5	35	0.23	1311	10	20.2	30.3	9.32
6	35	0.23	1318	10	20.4	32	8.12
7	35	0.28	1304	10	21.7	33.8	9.58
8	35	0.28	1311	10	21.1	29.3	9.27
9	35	0.28	1318	10	21.4	31	8.09
10	55	0.18	1304	10	18.8	30.2	9.24
11	55	0.18	1311	10	18.3	27	9.13
12	55	0.18	1318	10	18.5	28.5	8.17
13	55	0.23	1304	10	20	29.5	9.11
14	55	0.23	1311	10	19.5	26.2	9.08
15	55	0.23	1318	10	19.8	27.9	8.16
16	55	0.28	1304	10	20.9	27.9	8.97
17	55	0.28	1311	10	20.4	24.7	9
18	55	0.28	1318	10	20.7	26.3	8.12
19	70	0.18	1304	10	17.9	24.5	8.86
20	70	0.18	1311	10	17.6	23	8.96
21	70	0.18	1318	10	18	25.1	8.22
22	70	0.23	1304	10	19.1	23.8	8.65
23	70	0.23	1311	10	18.8	22	8.82
24	70	0.23	1318	10	19.1	24	8.13
25	70	0.28	1304	10	19.9	22.1	8.46
26	70	0.28	1311	10	19.6	20.4	8.69
27	70	0.28	1318	10	19.9	22.2	8.05

28	80	0.18	1304	10	16.9	19.5	8.62
29	80	0.18	1311	10	16.8	19.2	8.8
30	80	0.18	1318	10	17.4	21.7	8.2
31	80	0.23	1304	10	18.1	19.1	8.33
32	80	0.23	1311	10	18	18.5	8.59
33	80	0.23	1318	10	18.5	20.7	8.04
34	80	0.28	1304	10	18.9	17.7	8.12
35	80	0.28	1311	10	18.7	16.9	8.42
36	80	0.28	1318	10	19.2	18.9	7.92